

Open Access Textbook: The Students' Perspective in an Undergraduate Mathematical Thinking Statistics Course with Service Learning

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An inclusive theoretical model: Integrating critical multicultural instructional design with service learning and adopting an open access text to maximize equity, access, and success for all learners. Increased mathematical literacy necessary for all to achieve effective global citizenship and participation.

The Educational Experience

Project Description

To create a climate and instructional space for non-mathematics majors to embrace mathematics and develop increased mathematics literacy, a model of critical multicultural instructional design with service learning was employed. Student perceptions (attitudes, behaviours, beliefs) of an open source text were collected through pre- and post- surveys and student journals before and after major assignments and tests. This poster highlights students' characteristics related to their attitudes towards open access text/materials, course outcomes, and the service learning project.

Research Question

What are students' attitudes and beliefs regarding the use of published vs. open-access textbook verses open-access textbooks for an undergraduate, liberal education, mathematics course? What is the impact of using an open-access textbook student learning (self-reported)?

Definitions

Mathematics Literacy: An individual's capacity to identify and understand the role that mathematics plays in the world, to make well-founded judgments and to use and engage with mathematics in ways that meet the needs of the individual's life as a constructive, concerned and reflective citizen. (OECD, 2006, p. 72)

Service Learning: A teaching and learning strategy that integrates meaningful community service with instruction and reflection to enrich the learning experience, teach civic responsibility, and strengthen communities. (http://www.servicelearning.org/what-service-learning)

Open Access (OA) Textbooks: Textbook that are digital, online, free of charge, peer-reviewed and free of most copyright and licensing restrictions. (Suber, June 2012)

Integrated Multicultural Instructional Design (IMID): An integration of multicultural content and diverse teaching and learning strategies in postsecondary curricula, programs, courses, and academic support services through a collaborative community of postsecondary administrators, faculty, student development and academic support professionals, and students who express their commitment to diversity, multiculturalism, and social justice throughout the educational process. (Hibgee & Goff, 2009)

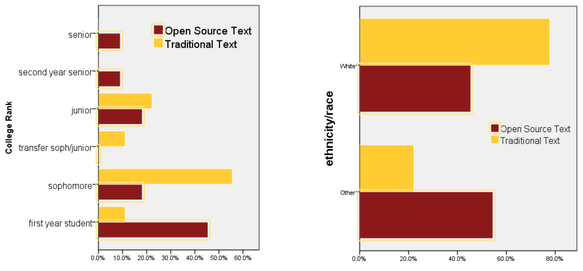
Investigation Methods

Context: An undergraduate course that fulfills the mathematical thinking liberal education graduation requirement. The course is also a required course for some nonmathematics-based undergraduate majors. Students are enrolled in Midwestern United States, tier one, doctoral research university. The hybrid introductory statistics course that is the subject of this presentation was revamped with the purpose of using a free open access book with open access applets for: (a) exploring statistical content/concept; (b) infusing a social justice theme; (c) using Microsoft Excel with shareware add-ins for statistical analysis; and (d) requiring students to participate with a non-governmental or non-profit organization in the design, implementation and analysis of data important to the agency. The on-campus sessions were conducted within a computer classroom. The off-campus sessions were conducted utilizing Moodle and engaging with service agencies.

Points of Evidence – Artifacts and Observation: Grades, journal entries, analysis of evidence of mathematical thinking and engagement within assignments and tests, within-semester and post-semester survey, comparisons between and among demographic and pre-enrollment characteristics. [open source text (n=11), traditional text (n=9)]

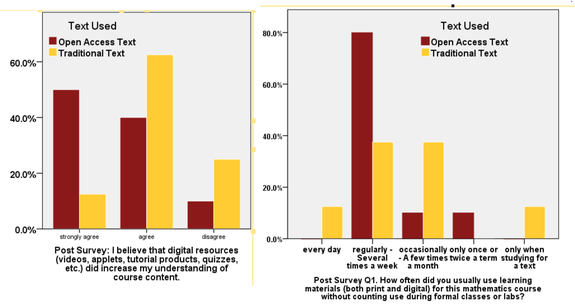
"Contrary to what I am used to, it [this class] tested our ability to use our knowledge and learning of stats to interpret data, as opposed to just plugging in our knowledge into calculations. This class encouraged me to explore how stats affects my life as well and the world around me. I was able to use my learning from our major project to interpret most of the information on the [major] tests. Unlike previous math classes I have taken, learning for this class did not require memorization of formulas but rather an interest in applied learning. Though it was hard for me to at first adjust to learning differently, I believe I learned more from this class than I expected to and really appreciate that."

Ranking and Ethnicity of Students (n=20) in the Fall 2013 Study



Preliminary Outcomes

- Reflective comments by students (when questions stem not specifying Service Learning or Open Source Text) generally reported on the value and challenge of service learning and infrequently commented on the Open Source Text or Traditional Text.
- There was a statically significant difference on the post survey between Open Source Text students (M=1.6, SD = .70) and Traditional Text students (M=2.5, SD=.76) on the question: I believe that digital resources did increase my understanding of the course content ; t(14.56)=-2.6, p=.021. (The scale was 1=strongly agree to 4=strongly disagree)
- There was a statically significant difference on the post survey between Open Source Text students (M=1.4, SD = .52) and Traditional Text students (M=2.25, SD=.46) on the question: I believe that open access textbooks provided the same level of quality as a traditional published print textbook ; t(15.74)=-3.68, p=.002. (The scale was 1=strongly agree to 4=strongly disagree)
- There was more consistency in student responses to frequency of use of learning materials with the Open Access Text.
- There were more non-white (60% non-w vs. 40% w) students, more first year college students and more female (65% f vs. 35% m) students who choose the Open Access Text.



"The process made me value civil service . . . it felt good to create such a thoughtful project for a non-profit. I also believe that the project helped me gain the knowledge of new statistical processes that I had never heard of. I never realized that math was this practical, and statistics truly is. I am thankful that I got the chance to work with such a great group on a challenging project. The skills I gained from this project will always be with me."

Implications for Practice and Next Steps

Implications for Practice:

- **Implementing IMID and Service Learning is demanding and necessitates negotiations with student throughout the semester to maximize engagement and real world learning.**
- **Making explicit the IMID model's process (including Service Learning) and anticipated outcomes is important to its successful implementation.**
- **The use of the Open Access Text added to students' engagement in the class and to goodwill "points" for the instructor. It also saved students up to $140 dollars/semester.**

Next Steps:

- **Develop analysis rubrics based that will be used to triangulate artifacts and observational data sources.**
- **Design a Professional Development Model in collaboration with the American Mathematical Association for Two-Year Colleges (AMATYC's Research in Mathematics Education for Two-Year Colleges [RMETYC] committee) to engage undergraduate mathematics faculty in classroom based research which uses artifacts and observational data sources to document learning and engagement outcomes.**

IMID Model Incorporating Undergraduate Mathematics Standards

How we learn / How we teach:

Establish and communicate clear expectations. Provide a welcoming learning environment. Work collaboratively to construct knowledge.
Identify what skills must be developed. Incorporate students' natural approaches to problem solving.
Integrate skill development with the acquisition of content knowledge. Incorporate critical thinking and meaning-making activities.
Promote understanding of how knowledge and personal experiences are shaped by context.
Understand that learning is a complex process involving many layers – engage students in metacognitive reflection and strategies.
Use teaching methods that consider diverse learning styles, abilities, ways of knowing, experiences and background knowledge.

What we learn / What we teach

Determine what content mastery is essential for each course and for the program or curriculum and establish corresponding course objectives that do not exclude students on the basis of gaps in prior knowledge.
Emphasize big mathematical ideas (concepts, ways of thinking, problem solving).
De-emphasize isolated, content-specific techniques, and components.
Interface mathematics with life-based questions that are relevant and important to the student population.
Incorporate problem solving, reasoning and proof, communication, connections and multiple representations within skill development
Integrate multicultural and global perspectives within course content.
Relate content to historical trends, current events, and future directions.

How we access academic support services / How we support learning

Use intrusive means to exposes students to one-on-one, small group, technology-based, and other academic support services.
Allow students multiple venues or provide multiple modes for "time-on-task" activities.
Provide services or opportunities that scaffold the learning process.
Use assignments and assessments as "candid snapshots" that can revised, revisited, or retaken.

How we demonstrate what we have learned / How we assess learning

Balance the weight of all assessment methods (formative and summative) so that all learning styles are supported and acknowledge when evaluating the essential components.
Develop multiple ways for students to demonstrate knowledge. mastery of course materials.
Use formative reflection journals for assessing "what is working" and "what is not working" throughout the semester.
Establish and communicate clear expectations in terms of learning objectives and engagement and evaluation measures in the teaching and learning process.

